

Exercise (SNM-CPA implies IND-CPA). Let $\mathfrak{C} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a public key cryptosystem. We say that the crypto-system is $(t, t_\pi, t_m, \varepsilon)$ -non-malleable against statically chosen relations (SNM-CPA) if for all t -time adversaries $\mathcal{A}$ the advantage

$$\text{Adv}_{\mathfrak{C}}^{\text{s-nm-cpa}}(\mathcal{A}) = |\Pr[\mathcal{G}_0^{\mathcal{A}} = 1] - \Pr[\mathcal{G}_1^{\mathcal{A}} = 1]| \leq \varepsilon$$

where the security games are defined as follows

$$\begin{array}{ll} \mathcal{G}_0^{\mathcal{A}} & \mathcal{G}_1^{\mathcal{A}} \\ \left[\begin{array}{l} \pi(\cdot) \leftarrow \mathcal{A} \\ (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}(\text{pk}) \\ m \leftarrow \mathcal{M}_0 \\ c \leftarrow \text{Enc}_{\text{pk}}(m) \\ \hat{c}_1, \dots, \hat{c}_n \leftarrow \mathcal{A}(c) \\ \text{if } c \in \{\hat{c}_1, \dots, \hat{c}_n\} \text{ then } \mathbf{return} \ 0 \\ \mathbf{return} \ \pi(m, \text{Dec}_{\text{sk}}(\hat{c}_1), \dots, \text{Dec}_{\text{sk}}(\hat{c}_n)) \end{array} \right. & \left[\begin{array}{l} \pi(\cdot) \leftarrow \mathcal{A} \\ (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}(\text{pk}) \\ m \leftarrow \mathcal{M}_0, \bar{m} \leftarrow \mathcal{M}_0 \\ \bar{c} \leftarrow \text{Enc}_{\text{pk}}(\bar{m}) \\ \hat{c}_1, \dots, \hat{c}_n \leftarrow \mathcal{A}(\bar{c}) \\ \text{if } \bar{c} \in \{\hat{c}_1, \dots, \hat{c}_n\} \text{ then } \mathbf{return} \ 0 \\ \mathbf{return} \ \pi(m, \text{Dec}_{\text{sk}}(\hat{c}_1), \dots, \text{Dec}_{\text{sk}}(\hat{c}_n)) \end{array} \right. \end{array}$$

and the sampling algorithm $\mathcal{M}_0$ is guaranteed to be a t_m -time algorithm and the predicate π is guaranteed to be t_π -time algorithm. Show that SNM-CPA security implies IND-CPA security by giving an explicit definition of the target relation π . Also, give the explicit quantification of running-time bounds. Interpret the results.

Solution. Let $\mathcal{B}$ be an adversary against IND-CPA. Then we construct the tree stage adversary $\mathcal{A}$. The first stage of the adversary $\mathcal{A}$ always outputs the binary relation $\pi(x, y) = [x = y]$ and the second and the third stage of the adversary are defined as follows

$$\begin{array}{ll} \mathcal{A}(\text{pk}) & \mathcal{A}(c) \\ \left[\begin{array}{l} (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ \mathcal{M}_0 = \{m_0, m_1\} \\ \mathbf{return} \ \mathcal{M}_0 \end{array} \right. & \left[\begin{array}{l} g \leftarrow \mathcal{B}(c) \\ \hat{c}_1 \leftarrow \text{Enc}_{\text{pk}}(m_g) \\ \mathbf{return} \ \hat{c}_1 \end{array} \right. \end{array}$$

where $\mathcal{M}_0$ denotes the uniform distribution over m_0 and m_1 .

As the predicate π is fixed, we can omit the predicate selection stage and replace the definition of the predicate into the games $\mathcal{G}_0$ and $\mathcal{G}_1$. As a result, we get the following simplification of the games:

$$\begin{array}{ll} \mathcal{G}_2^{\mathcal{A}} & \mathcal{G}_3^{\mathcal{A}} \\ \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}(\text{pk}) \\ m \leftarrow \mathcal{M}_0 \\ c \leftarrow \text{Enc}_{\text{pk}}(m) \\ \hat{c}_1, \dots, \hat{c}_n \leftarrow \mathcal{A}(c) \\ \text{if } c \in \{\hat{c}_1, \dots, \hat{c}_n\} \text{ then } \mathbf{return} \ 0 \\ \mathbf{return} \ [m \neq \text{Dec}_{\text{sk}}(\hat{c}_1)] \end{array} \right. & \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}(\text{pk}) \\ m \leftarrow \mathcal{M}_0, \bar{m} \leftarrow \mathcal{M}_0 \\ \bar{c} \leftarrow \text{Enc}_{\text{pk}}(\bar{m}) \\ \hat{c}_1, \dots, \hat{c}_n \leftarrow \mathcal{A}(\bar{c}) \\ \text{if } \bar{c} \in \{\hat{c}_1, \dots, \hat{c}_n\} \text{ then } \mathbf{return} \ 0 \\ \mathbf{return} \ [m \neq \text{Dec}_{\text{sk}}(\hat{c}_1)] \end{array} \right. \end{array}$$

As the second stage of $\mathcal{A}$ returns $\mathcal{M}_0$ that is a uniform distribution over $\{m_0, m_1\}$, we can replace sample generation by uniform choice of index bits $i, j \leftarrow \{0, 1\}$. We can also replace the query $\mathcal{A}(c)$ by the explicit

description of the third stage of the adversary. This leads us to the following games:

$$\begin{array}{ll}
\mathcal{G}_4^{\mathcal{B}} & \mathcal{G}_5^{\mathcal{B}} \\
\left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\} \\ c \leftarrow \text{Enc}_{\text{pk}}(m_i) \\ g \leftarrow B(c) \\ \hat{c} \leftarrow \text{Enc}_{\text{pk}}(m_g) \\ \text{if } c = \hat{c} \text{ then return } 0 \\ \text{return } [m_i \neq \text{Dec}_{\text{sk}}(\hat{c})] \end{array} \right. & \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\}, j \leftarrow \{0, 1\} \\ \bar{c} \leftarrow \text{Enc}_{\text{pk}}(m_j) \\ g \leftarrow B(c) \\ \hat{c} \leftarrow \text{Enc}_{\text{pk}}(m_g) \\ \text{if } \bar{c} = \hat{c} \text{ then return } 0 \\ \text{return } [m_i \neq \text{Dec}_{\text{sk}}(\hat{c})] \end{array} \right.
\end{array}$$

We can further simplify the games, as $\text{Dec}_{\text{sk}}(\hat{c}) = m_g$ by the perfect functionality of the cryptosystem:

$$\begin{array}{ll}
\mathcal{G}_6^{\mathcal{B}} & \mathcal{G}_7^{\mathcal{B}} \\
\left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\} \\ c \leftarrow \text{Enc}_{\text{pk}}(m_i) \\ g \leftarrow B(c) \\ \text{if } c = \text{Enc}_{\text{pk}}(m_g) \text{ then return } 0 \\ \text{return } [m_i \neq m_g] \end{array} \right. & \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\}, j \leftarrow \{0, 1\} \\ \bar{c} \leftarrow \text{Enc}_{\text{pk}}(m_j) \\ g \leftarrow B(\bar{c}) \\ \text{if } \bar{c} = \text{Enc}_{\text{pk}}(m_g) \text{ then return } 0 \\ \text{return } [m_i \neq m_g] \end{array} \right.
\end{array}$$

As the if branch makes the analysis of the games more difficult, we first analyse the without these branches. This a good approximation. A cryptosystem can be IND-CPA secure only if such ciphertext collisions occur with a negligible probability. We will later analyse this issue in further.

$$\begin{array}{ll}
\mathcal{G}_8^{\mathcal{B}} & \mathcal{G}_9^{\mathcal{B}} \\
\left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\} \\ c \leftarrow \text{Enc}_{\text{pk}}(m_i) \\ g \leftarrow B(c) \\ \text{return } [m_i \neq m_g] \end{array} \right. & \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\}, j \leftarrow \{0, 1\} \\ \bar{c} \leftarrow \text{Enc}_{\text{pk}}(m_j) \\ g \leftarrow B(\bar{c}) \\ \text{return } [m_i \neq m_g] \end{array} \right.
\end{array}$$

Since i is not used before the last line in the game $\mathcal{G}_9$, we can rewrite the game as follows:

$$\mathcal{G}_9^{\mathcal{B}} \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ j \leftarrow \{0, 1\} \\ \bar{c} \leftarrow \text{Enc}_{\text{pk}}(m_j) \\ g \leftarrow B(\bar{c}) \\ i \leftarrow \{0, 1\} \\ \text{return } [m_i \neq m_g] \end{array} \right.$$

From this game construction it is very tempting to say that $\Pr[\mathcal{G}_9^{\mathcal{B}} = 1] = \frac{1}{2}$. However, this is not true! For example consider an adversary $\mathcal{A}$ that always outputs $m_0 = m_1$. Then clearly $\Pr[\mathcal{G}_9^{\mathcal{B}} = 1] = 1$.

Intuitively, an adversary $\mathcal{B}$ that return $m_0 = m_1$ with high probability is not useful, as this event diminishes their success against the IND-CPA games. However, we must prove this formally. Let us consider the advantage of $\mathcal{B}$ against IND-CPA games:

$$\begin{array}{ll} \mathcal{Q}_0^{\mathcal{B}} & \mathcal{Q}_0^{\mathcal{B}} \\ \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ c \leftarrow \text{Enc}_{\text{pk}}(m_0) \\ \text{return } \mathcal{B}(c) \end{array} \right. & \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ c \leftarrow \text{Enc}_{\text{pk}}(m_1) \\ \text{return } \mathcal{B}(c) \end{array} \right. \end{array}$$

Now we can express the success of $\mathcal{B}$ against IND-CPA games by exhaustive decomposition:

$$\begin{aligned} \Pr [\mathcal{Q}_0^{\mathcal{B}} = 1] &= \Pr [\mathcal{Q}_0^{\mathcal{B}} = 1 \wedge m_0 = m_1] + \Pr [\mathcal{Q}_0^{\mathcal{B}} = 1 \wedge m_0 \neq m_1] \\ \Pr [\mathcal{Q}_1^{\mathcal{B}} = 1] &= \Pr [\mathcal{Q}_1^{\mathcal{B}} = 1 \wedge m_0 = m_1] + \Pr [\mathcal{Q}_1^{\mathcal{B}} = 1 \wedge m_0 \neq m_1] \end{aligned}$$

If $m_0 = m_1$ then the ciphertext in both games has identical distribution and thus we can prove

$$\Pr [\mathcal{Q}_0^{\mathcal{B}} = 1 \wedge m_0 = m_1] = \Pr [\mathcal{Q}_1^{\mathcal{B}} = 1 \wedge m_0 = m_1] .$$

As a consequence,

$$\text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B}) = |\Pr [\mathcal{Q}_0^{\mathcal{B}} = 1 \wedge m_0 \neq m_1] - \Pr [\mathcal{Q}_1^{\mathcal{B}} = 1 \wedge m_0 \neq m_1]| .$$

There are now two ways how to proceed with the proof. First, we could do the same decomposition for the games $\mathcal{G}_8$ and $\mathcal{G}_9$. However, this would lead us to the lengthy probability calculations. Instead, we use a shortcut. Note that given $\mathcal{B}$ which can occasionally output $m_0 = m_1$, we can define $\mathcal{B}^\circ$ that runs $\mathcal{B}$ and filters out these unfortunate events:

$$\begin{array}{ll} \mathcal{B}^\circ(\text{pk}) & \mathcal{B}^\circ(c) \\ \left[\begin{array}{l} (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ \text{if } (m_0 = m_1) \text{ then return } (m_0^*, m_1^*) \\ \text{else return } (m_0, m_1) \end{array} \right. & \left[\begin{array}{l} \text{if } (m_0 = m_1) \text{ then return } 0 \\ \text{else return } \mathcal{B}(c) . \end{array} \right. \end{array}$$

Then clearly $\text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B}^\circ) = \text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B} \circ)$ and the running times differ by a constant.

Hence, we can assume that $\mathcal{B}$ never returns $m_0 = m_1$ and proceed further with the analysis. Under the assumption $m_0 \neq m_1$, the condition $m_i \neq m_g$ is equivalent to the condition $i \neq g$ and we can indeed conclude that $\Pr [\mathcal{G}_9^{\mathcal{B}} = 1] = \frac{1}{2}$. The game $\mathcal{G}_8$ simplifies to

$$\begin{array}{l} \mathcal{G}_8^{\mathcal{B}} \\ \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ (m_0, m_1) \leftarrow \mathcal{B}(\text{pk}) \\ i \leftarrow \{0, 1\} \\ c \leftarrow \text{Enc}_{\text{pk}}(m_i) \\ g \leftarrow B(c) \\ \text{return } [i \neq g] , \end{array} \right. \end{array}$$

for which we know from previous exercises that $\Pr [\mathcal{G}_8^{\mathcal{B}} = 1] = \frac{1}{2} \pm \frac{1}{2} \cdot \text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B})$. Thus, we have proved

$$|\Pr [\mathcal{G}_8^{\mathcal{B}} = 1] - \Pr [\mathcal{G}_9^{\mathcal{B}} = 1]| = \frac{1}{2} \cdot \text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B}) .$$

To complete the proof, we must relate this difference to the difference between the games $\mathcal{G}_6$ and $\mathcal{G}_7$. First note that if $c = \text{Enc}_{\text{pk}}(m_g)$ then $m_i = m_g$ and the adversary cannot win the game $\mathcal{G}_8$. Thus, we have proved

$$\Pr [\mathcal{G}_6^{\mathcal{B}} = 1] = \Pr [\mathcal{G}_8^{\mathcal{B}} = 1] .$$

For the game $\mathcal{G}_7$, we can conclude only that

$$\Pr [\mathcal{G}_7^{\mathcal{B}} = 1] \leq \Pr [\mathcal{G}_9^{\mathcal{B}}] \leq \frac{1}{2} .$$

Without loss of generality we can assume that the $\mathcal{B}$ is such that

$$\Pr [\mathcal{G}_8^{\mathcal{B}} = 1] \geq \frac{1}{2} .$$

If the inequality does not hold we can invert the the outcome of $\mathcal{B}$ to make it hold. The running time will increase only by constant. Now combining all inequalities we obtain

$$\Pr [\mathcal{G}_6^{\mathcal{B}} = 1] - \Pr [\mathcal{G}_7^{\mathcal{B}} = 1] \geq \Pr [\mathcal{G}_8^{\mathcal{B}} = 1] - \frac{1}{2} = |\Pr [\mathcal{G}_8^{\mathcal{B}} = 1] - \Pr [\mathcal{G}_9^{\mathcal{B}} = 1]| = \frac{1}{2} \cdot \text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B}) .$$

Hence, we have proved that

$$\text{Adv}_{\mathcal{E}}^{\text{s-nm-cpa}}(\mathcal{A}) \geq \frac{1}{2} \cdot \text{Adv}_{\mathcal{E}}^{\text{ind-cpa}}(\mathcal{B})$$

for the adversary $\mathcal{A}$ for which the running time is only constant times larger than the running time of $\mathcal{B}$. Thus, $(t, t_{\pi}, t_m, \varepsilon)$ -non-malleability against statically chosen relations implies $(t, 2\varepsilon)$ -IND-CPA security.